NNN-R-770
April 7, 1966

SUPERSEDING
Fed. Spec. GGG-R-810
August 11, 1953

FEDERAL SPECIFICATION

RULE, LABORATORY (MEASURING, PLASTIC AND WOOD)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic rules and wood meter sticks for general linear measurements.

1.2 Classification. Rules shall be of the following classes and sizes, as specified (see 6.2):

Class 1—Plastic.
Size—6-inch.
Size—12-inch.
Size—18-inch.

Class 2—Wood.
Size—1-meter.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

ZZ-S-751—Stoppers, Bottle, Rubber.
LLL-S-731—Stopper, Bottle, Cork.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Class 1, plastic rules.* These rules shall be of a high quality copolymer of vinyl chloride and vinyl acetate. The material shall be white in color, unless otherwise specified (see 6.2).

3.1.2 *Class 2, wood rules.* These rules

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shall be of well-seasoned hardwood of either of the following species: birch, boxwood, or hard maple. The wood shall be straight-grained and clear throughout.

3.2 **Finish.**

3.2.1 *Class 1, plastic rules.* All surfaces of these rules shall be smooth.

3.2.2 *Class 2, wood rules.* Wood rules shall be smoothly finished all over with one or more coats of either white shellac, clear lacquer, or clear spar varnish. The coating shall be such as to provide adequate protection under normal laboratory conditions.

3.3 **Identification.** Each rule shall be plainly and permanently marked with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. When specified (see 6.2), rules shall be marked with manufacturers' part number.

3.4 **Graduations.**

3.4.1 *Dimensions and accuracy of lines.* Graduation lines shall be straight, free from discontinuities and ragged edges, and shall be not more than 0.01 inch in width. Graduation lines for corresponding divisions shall be uniform and of lengths conforming to good commercial practice. For plastic rules, when tested in accordance with 4.4.1 the scale error between any two graduations 2 inches or 5 centimeters apart shall not exceed 0.01 inch or 0.25 millimeter (mm.), and the error between end graduations shall not exceed 0.015 inch, 0.020 inch, and 0.025 inch or 0.4 mm., 0.5 mm., and 0.6 mm., respectively, for the 6-, 12-, and 18-inch sizes. For wood rules, the scale error between any two graduations 2 inches or 5 centimeters apart shall not exceed 0.040 inch or 1 mm. Stopper size and temperature scale graduations, when provided, shall conform to corresponding accuracy requirements.

3.4.2 *English system.* Rules with inch scales shall be graduated in inches to sixteenths of an inch for plastic rules and inches to eighths of an inch for wood rules. Each inch graduation shall be plainly and distinctly marked with the corresponding Arabic numeral. The inch graduations shall be relatively long, the half-inch graduations of medium length, and those of the lesser fractions proportionately shorter.

3.4.3 *Metric scales.* Rules with metric scales shall be graduated in centimeters and millimeters. Each centimeter graduation shall be plainly and distinctly marked with the corresponding Arabic numeral. The centimeter graduations shall be relatively long, the half-centimeter graduations of medium length, and the millimeter graduations relatively short.

3.4.4 *Stopper-size and temperature-scale graduations.* Stopper-size scales shall indicate top diameters of cork stoppers in sizes 000 to 26, inclusive, and of rubber stoppers in sizes 00 to 13, inclusive, as given in LLL-S-731 and ZZ-S-751. Each stopper-size scale shall show sizes progressively from its zero reference line, and all size graduations shall be plainly and distinctly marked with the corresponding Arabic numerals. Temperature scales shall consist of the F. and C. scales laid out so as to provide direct comparison between the two scales. The graduations shall cover the temperature range represented by the values 212° and —10°F. Each scale shall be plainly and distinctly marked and graduated to indicate the name of the scale, the scale degrees, and scale readings in corresponding Arabic numerals at intervals of not more than ten degrees.

3.4.5 *Scale identifications.* The words "inches" and "centimeters" shall appear in upper-case letters near the graduation lines at the beginning of the scale. When rulers are specified with stopper-size and temperature scales on reverse side, the words "corks" and "rubber stoppers" shall appear in upper-case letters near the scale. In addition, the legend "size refers to top diameters" shall appear in upper-case letters near to the scale.

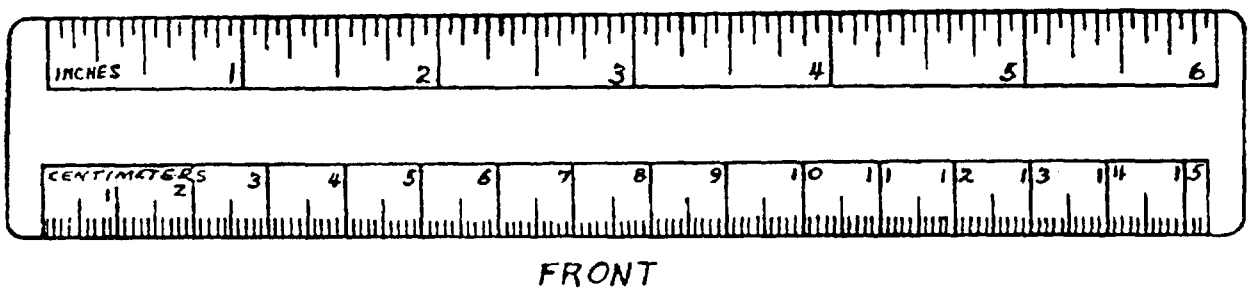
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3.5 Class 1, plastic rules. Plastic rules shall be similar to figure 1 and comply with the applicable dimensions specified in table I.

TABLE I. *Dimensions of class 1, plastic rules*

Size	Scale length[1]		Overall length maximum	Width		Thickness	
	English	Metric		Minimum	Maximum	Minimum	Maximum
Inches	*Inches*	*mm.*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*
6	6	152	6⅜	⅞	1¼	0.030	0.040
12	12	305	12⅜	1⅜	1⅝	.030	.040
18	18	457	18½	1¾	2⅛	.030	.040

[1] For tolerances see 3.4.1.



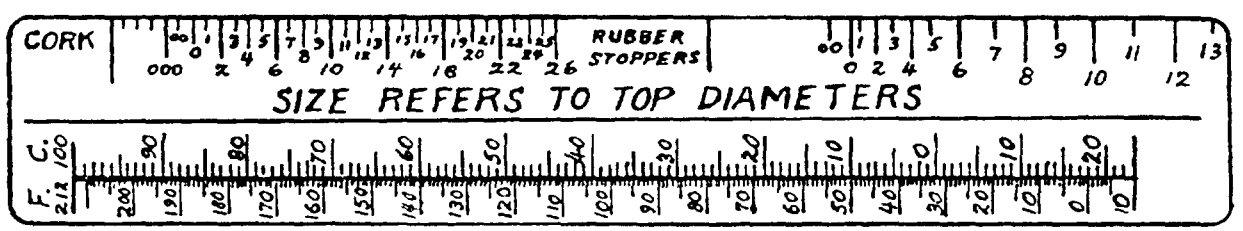


FIGURE 1. Class 1, plastic

3.5.1 *Construction and dimensions.* Plastic rules shall be so constructed as to lie flat on a plane surface and shall be sufficiently flexible to meet the requirements of 4.4.2. Unless otherwise specified (see 6.2), rules shall extend at least 1/8 inch beyond the graduations at each end. Ends so extended shall be square with the edge faces of the rule and shall have well-rounded corners. When rules with overall lengths equal to the lengths of their scales are specified, both ends shall be square with the edge faces, and the corners shall be square. When rules with one extended end are specified, both ends shall be square with the edge faces, the corners of the extended end shall be rounded, and the corners of the opposite end shall be square.

3.5.2 *Graduation marking.* Graduation lines shall be either stamped and filled with permanent black ink or printed with permanent black ink. Each rule shall be provided with both the English and metric scale, located at opposite edges of the same face. Unless otherwise specified (see 6.2), the two scales shall read from left to right and from the same end. Both scales of rules with one extended end shall read from the end not extended. When specified (see 6.2),

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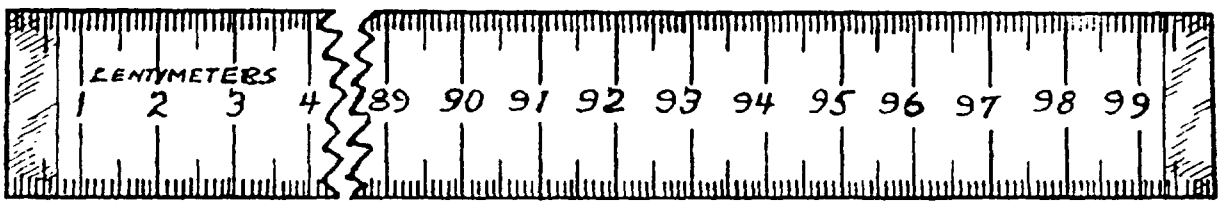
the reverse face of the rule shall have the stopper-size scales along one edge and the temperature scales along the other edge.

3.6 Class 2, wood rules. Wood rules shall be similar to figure 2 and comply with the dimensions specified in table II.

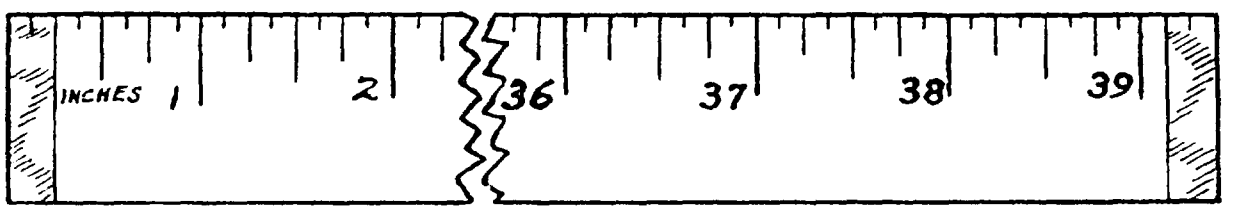
TABLE II. *Dimensions of class 2, wood rules*

Size	Scale length[1]		Width		Thickness	
	Metric	English	Minimum	Maximum	Minimum	Maximum
	cm.	*Inches*	*mm.*	*mm.*	*mm.*	*mm.*
1 meter	100	39.37	25	31	7	8.75

[1] For tolerances see 3.4.1.



FRONT



BACK

FIGURE 2. Class 2, wood

3.6.1 *Construction and dimensions.* Each rule shall be of one piece of natural wood, and shall be capable of resisting warpage as required when tested in accordance with 4.4.3.2. Each rule shall have a metal tip or cap at each end, flush with the main and edge faces of the body of the rule. The tips or caps shall be of brass or highly corrosion-resistant steel, and each shall be attached by two or more steel or brass throughpins (rivets), finished flush with the surfaces of the tips or caps.

3.6.2 *Graduation markings.* Graduation lines shall be either stamped or cut, and filled with permanent black ink. One face shall be graduated along both edges in the metric system, both scales reading from the same end. The opposite face shall be graduated along one edge in the English system. All scales shall read from left to right, unless otherwise specified (see 6.2).

3.7 Workmanship. Workmanship shall be first class throughout. Rules shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless

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otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sample.

4.2.1 *Lot.* All rules of the same class and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 *Sampling for quality conformance.* A random sample rule shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. Each sample rule selected shall be tested in accordance with 4.4 as applicable, to verify compliance with this specification. Any sample rule which does not conform to the requirements shall be rejected. If the number of nonconforming rules in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Quality conformance inspection.** Quality conformance inspection shall be applied to each sample selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(a) Examination—4.3.1.
(b) Tests—4.3.2.
(c) Examination of preservation, packaging, packing, and marking for shipment and storage—4.5.

4.3.1 *Examination.* Each sample rule selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table III (with AQL as indicated). Any rule in the sample containing one or more defects shall not be offered for delivery. If the number of defective rules in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

TABLE III. *Classification of defects in accordance with MIL-STD-105*

Critical defects:	(None defined)
Major defects:	(AQL 1.5 percent defective)
101	Class and size of rule not as specified.
102	Rule not smooth or not flat.
103	Rule not one piece, not the required wood species, not clear and straight grained, or rule not coated as required (class 2 only).
104	Ends of rule not capped, cap not flush, caps not securely attached, rivets not finished flush, or cap material not as specified (class 2 only).
105	Graduations or numerals missing, illegible, ragged, not continuous, not uniform, or not as specified.
106	Scales not identified or scale identification illegible, incorrect, incomplete, or not as specified.
107	Scales (front and back) do not read from the same end of the rule or scales do not read from left to right.
108	Ends of rule (if not extended beyond the scale ends) not square with edge faces or corners not sharp (class 1 only).
Minor defects:	(AQL 2.5 percent defective)
201	Overall length of rule exceeds the specified maximum (class 1 only).
202	Width of rule not within the specified maximum and minimum.
203	Thickness of rule not within the specified maximum and minimum values.
204	Amount of rule extension (each end) less than the specified minimum (class 1 only).
205	Color of rule not as specified (class 1 only).
206	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.3.2 *Tests.* Each sample rule selected in accordance with 4.2.2 shall be tested as specified in 4.4.1 through 4.4.3.2 as applicable, with AQL to be 1.5 percent defective.

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4.4 Test procedures.

4.4.1 *Accuracy of graduated scales.* Sample rules shall be checked against calibrated master scales of known accuracy at room temperature (68°F.), and the results of such measurements shall fall within the limits specified in 3.4.1.

4.4.2 *Flexibility of class* 1, *plastic rules.* Sample rules shall be placed in a water bath and held at a temperature of 350° ± 2°F. for at least 30 minutes. Upon removal from the bath, the rules shall be wiped dry and immediately bent at least 180° around a round mandrel having a diameter of 1-1/2 inches. The rules shall not break, crack, or otherwise become damaged as a result of this test.

4.4.3 *Linear expansion.*

4.4.3.1 *Class* 1, *plastic rules.* Sample class 1, plastic rules shall be checked against a master scale and then be placed in a water bath and held at a temperature of 120° ± 2°F. for at least 30 minutes.

Upon removal from the bath, the rules shall be wiped dry and immediately checked against a master scale which has been held at room temperature (68°F.). The linear expansion shall not exceed the tolerances specified in 3.4.1 for the size of rule tested.

4.4.3.2 *Class* 2, *wood rules.* Sample class 2, wood rules shall be subjected to a relative humidity of 90 ± 2 percent at 70° ± 3°F. for not less than 48 hours, after which the rule shall be placed on a flat surface edgewise and subsequently flatwise. The maximum bow shall not exceed 1/16 and 1/8 inch, respectively (see 3.6.1).

4.5 Examination of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation, packaging, and packing shall be as specified (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Rules, as furnished under this specification, are for general laboratory use in linear measurement with the unaided eye.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Class and size (see 1.2).

(*c*) Material and color, if other than specified (see 3.1.1).

(*d*) Special marking, when required (see 3.3).

(*e*) Construction and dimensions, if other than specified (see 3.5.1).

(*f*) Scales required and location, if other than specified (see 3.5.2 and 3.6.2).

(*g*) Levels of preservation, packaging, and packing required (see 5.1).

6.3 The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of rules which are otherwise in accordance with the requirement of this specification.

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